

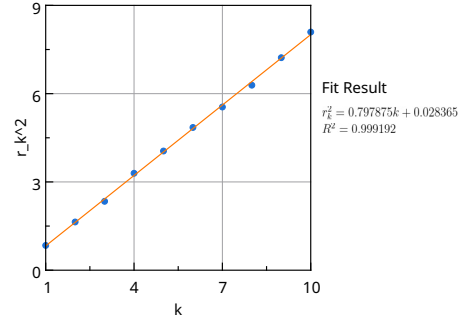
NewtonRings Experiment Report

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Newton's Rings Experimental Data

k	x_k (mm)	x'_k (mm)	r_k (mm)	r_k^2 (mm ²)
1	30.005	28.17	0.9174999999999999	0.8418062499999998
2	30.355	27.795	1.28	1.6384
3	30.61	27.55	1.53	2.3409
4	30.89	27.26	1.815	3.294225
5	31.11	27.085	2.0125	4.05015625
6	31.275	26.87	2.2025	4.85100625
7	31.42	26.71	2.355	5.546025
8	31.605	26.59	2.5075	6.28755625
9	31.775	26.4	2.6875	7.22265625
10	31.89	26.2	2.845	8.094025

Linear Fit



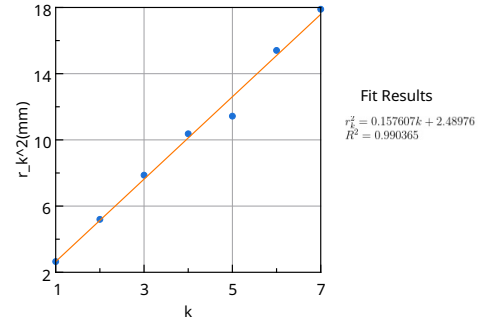
Linear fit of r_k^2 versus k

$$R = \frac{\alpha}{\lambda_{\text{Na}}} = \frac{0.797875 \text{ mm}^2}{5.893 \times 10^{-4} \text{ mm}} = 1.354 \times 10^3 \text{ mm} = 1.354 \text{ m}$$

LED Experimental Data

k	x_k (mm)	x'_k (mm)	r_k (mm)	r_k^2 (mm ²)
1	29.965	28.3375	1.6275	2.64875625
2	30.28	28	2.28	5.19840000000001
3	30.5325	27.7275	2.805	7.868025
4	30.73	27.51	3.22	10.3684
5	30.8925	27.511	3.3815	11.43454225
6	31.0625	27.1375	3.925	15.405625
7	31.205	26.975	4.23	17.8929

Linear Fit



Linear fit of r_k^2 versus k

$$\lambda_{\text{LED}} = \frac{\tilde{\alpha}}{\alpha} \lambda_{\text{Na}} = 468.3 \text{ nm}$$

Detailed data and figures are available at:

<https://github.com/alizeisnotalice/physics-experiment/NewtonRings>